Understanding Your
NEXT GENERATION SEQUENCING (NGS) Report
Quick Tips Care Guide

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Understanding About Treatment Planning

Next-Generation Sequencing (NGS) has fundamentally changed modern biology and medicine. This technology allows for the rapid and large-scale reading of genetic blueprints, such as DNA or RNA. NGS provides scientists and clinicians with unprecedented access to genomic information. The speed, scale, and affordability of this approach have made complex genomic studies routine, fueling breakthroughs in understanding human health, disease, and the diversity of life.
- BiologyInsights.com

Understanding the molecular profile of an LMS tumor may not provide all the answers today, but it is becoming an increasingly important part of informed treatment decision-making and clinical trial participation.

Does my tumor's genetic profile suggest that standard chemotherapy may work differently for me?
Patients might ask: "Does anything in my NGS report suggest I may respond better or worse to certain treatments?"

Are there treatment options that may be especially relevant because of my tumor's genetic findings?
Examples:

- Targeted therapies
- Combination therapies
- Clinical trials
- Immunotherapy studies
- Biomarker-driven studies

About Clinical Trials

Are there any clinical trials specifically designed for patients with my genetic profile?
This is becoming increasingly important as trials move toward:

- Biomarker-based enrollment
- Precision medicine approaches
- Molecularly selected patient groups

If my disease progresses, would my tumor genetics help identify future treatment options?
Patients are surprised to learn that NGS results may become more useful later in the disease course as new trials open.

About Prognosis

Does anything in my pathology report or NGS testing suggest my tumor behaves more aggressively or less aggressively?

The study identified factors such as:

- **High mitotic rate** – how quickly cells are dividing in the tumor: A higher mitotic count is directly correlated with a more aggressive tumor biology, suggesting faster growth and a greater likelihood of spreading
- **Tumor necrosis** – cell death with a tumor; influences disease progression
- **TP53 alterations** - The TP53 mutation is one of the most studied genetic alterations in cancer research, playing a critical role in tumor development, progression, and response to treatment.
- **RB1 alterations** (RB1 acts like a brake pedal on cell growth)

Patients often appreciate understanding how physicians interpret these findings.

Questions for Newly Diagnosed Patients

- What subtype of LMS do I have?
- Have we performed NGS testing?
- Should my tumor be tested now or saved for future testing?
- Are there biomarkers that could influence treatment decisions?
- Should I seek a sarcoma center second opinion regarding genomic findings?

Questions for Metastatic LMS Patients

- Should my tumor be retested if it was sequenced years ago?
- Are there new biomarkers or mutations that could affect treatment choices?
- Are there molecularly targeted trials available for me?
- Should circulating tumor DNA (liquid biopsy) be considered?

Resource Information

National LMS Foundation

Sarcoma & LMS Pathways

Care Guide - Quick Tips for Patients & Families

Trusted guidance for navigating diagnosis, treatment, and leiomyosarcoma

www.NLMSF.org